Heart of Algebra

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

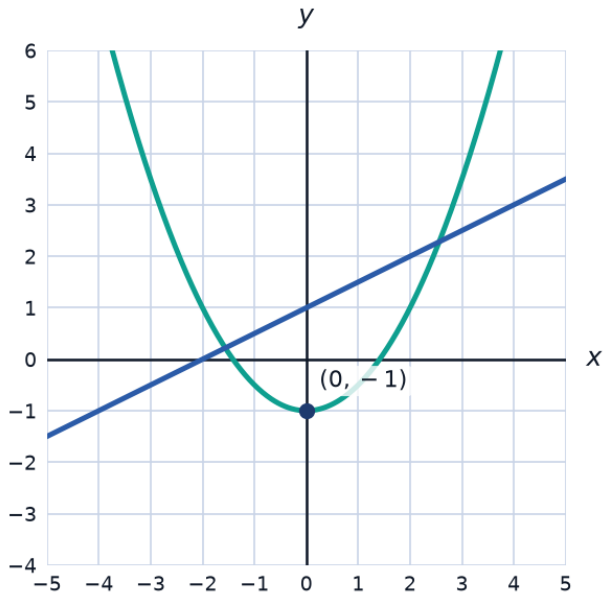
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving linear equations

Solve for x: $4x - 9 = 23$.

2 [Foundational] • Target: Solving linear equations

Solve for x: $\frac{3x+2}{5} = 4$.



3 [Foundational] • Target: Solving linear equations

Solve for x: $6(x - 3) = 2(x + 9)$.

4 [Foundational] • Target: Linear inequalities

Solve $5x - 3 > 12$.

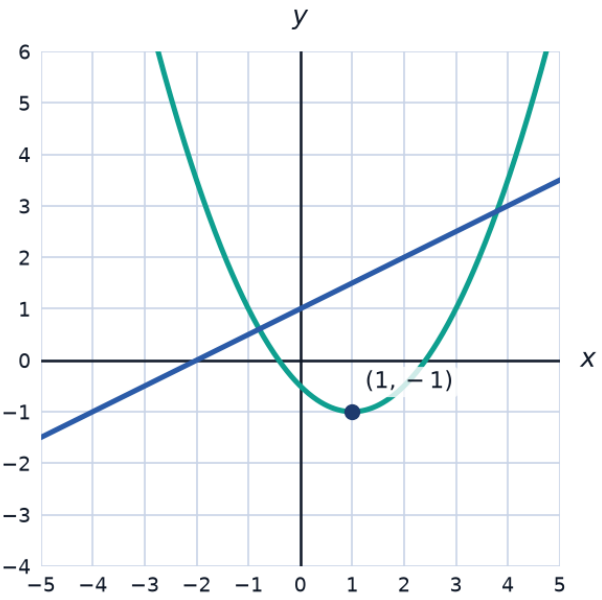
5 [Foundational] • Target: Linear inequalities

Solve $-2x + 7 \leq 15$.

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6 [Foundational] • Target: Linear inequalities

Solve $3(x-2) \geq x+4$.



7 [Foundational] • Target: Slope and equations of lines

Find the slope of the line through $(2, 5)$ and $(6, 13)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Slope and equations of lines

Write the equation, in slope-intercept form, of the line with slope -3 passing through $(0, 7)$.

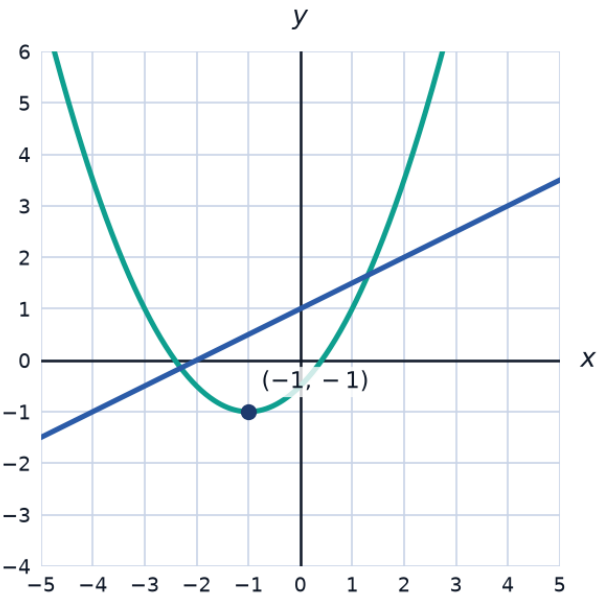
9 [Intermediate] • Target: Slope and equations of lines

Find the slope of a line perpendicular to $y = \frac{2}{5}x - 1$.

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10 [Intermediate] • Target: Systems of equations by substitution

Solve the system: $y = 3x - 4$ and $2x + y = 11$.



11 [Intermediate] • Target: Systems of equations by substitution

Solve the system: $x - 2y = 1$ and $3x + y = 17$.

12 [Intermediate] • Target: Systems of equations by elimination

Solve the system: $3x + 2y = 19$ and $x - 2y = 1$, using elimination.

13 [Intermediate] • Target: Systems of equations by elimination

Solve the system: $4x - y = 10$ and $2x + y = 14$, using elimination.

14 [Intermediate] • Target: Absolute value equations

Solve $|2x - 5| = 9$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

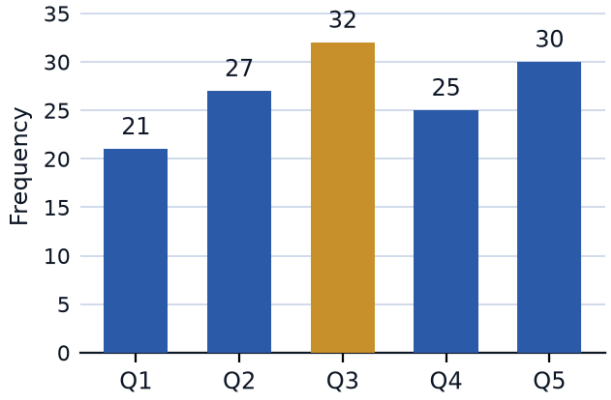
15 [Challenging] • Target: Absolute value equations • Cross-Lesson: 2.01 Quadratics & Polynomials

Solve $|x + 4| = 10$.

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16 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 3.01 Functions & Graphs

A phone plan costs $C = 20 + 0.05m$ dollars for m minutes of calls. Interpret the meaning of the numbers 20 and 0.05 in this context.



17 [Challenging] • Target: Interpreting linear models • Cross-Lesson: 4.01 Ratios, Rates & Data

Company A charges $C = 30 + 0.20d$ and Company B charges $C = 0.35d$ for a d-mile trip. Find the mileage at which both companies charge the same amount, and state which company is cheaper for a 150-mile trip.

18 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 5.01 Statistics & Probability

For what value of k does the equation $6x + 4 = kx + 4$ have infinitely many solutions?

19 [Challenging] • Target: Systems with no solution or infinitely many solutions • Cross-Lesson: 6.01 Geometry

Explain why the system $2x - 3y = 6$ and $4x - 6y = 10$ has no solution.

20 [Challenging] • Target: Compound inequalities • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Solve the compound inequality $-3 < 2x + 1 \leq 9$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving linear equations

Solve for x: $-2x + 19 = 23$.

a. -4

b. 0

c. -2

d. 2

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22 [Foundational] • Target: Linear inequalities

Solve for x: $2x + 11 = 13$.

a. -1

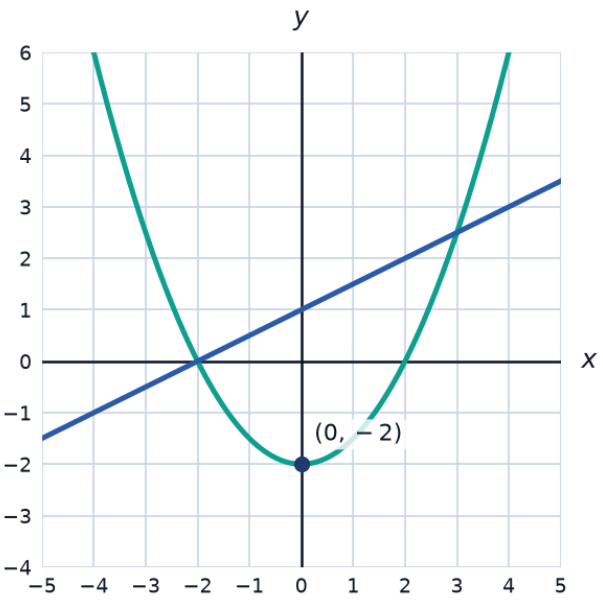
b. 3

c. 4

d. 1

23 [Foundational] • Target: Slope and equations of lines

Solve for x: $4x - 14 = 2$.



a. 6

b. -4

c. 4

d. 2

24 [Foundational] • Target: Systems of equations by substitution

Solve for x: $6x - 2 = 16$.

a. 5

b. 3

c. -3

d. 1

25 [Foundational] • Target: Systems of equations by elimination

Solve for x: $-8x + 1 = 9$.

a. 2

b. -3

c. 1

d. -1

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26 [Foundational] • Target: Absolute value equations

Solve for x: $8x - 4 = -124$.

a. -17

b. 15

c. -15

d. -13

27 [Foundational] • Target: Interpreting linear models

Solve for x: $4x - 11 = -15$.

a. 2

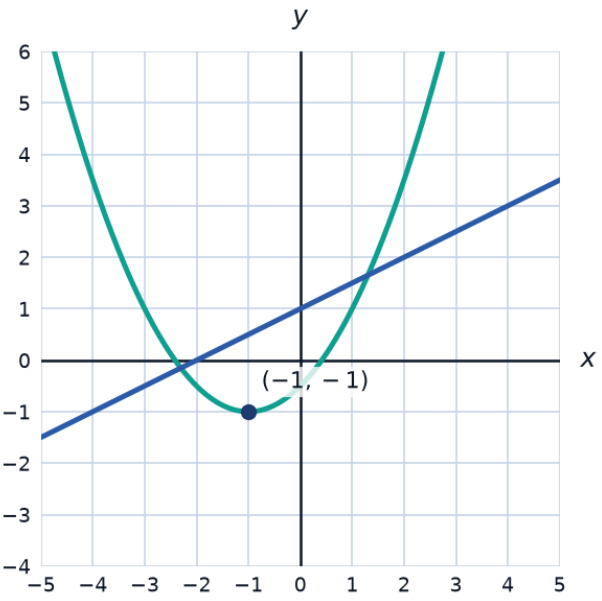
b. 1

c. -1

d. -3

28 [Foundational] • Target: Systems with no solution or infinitely many solutions

Solve the system: $4x + y = -14$ and $4x - y = 6$, using elimination.



a. $(-1, -12)$

b. $(-1, -10)$

c. $(1, -10)$

d. $(-10, -1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Compound inequalities

Solve the system: $6x + 4y = -44$ and $6x - 4y = 20$, using elimination.

a. $(-2, -10)$

b. $(-8, -2)$

c. $(-2, -8)$

d. $(0, -8)$

30 [Intermediate] • Target: Multi-step word problems

Solve the system: $2x + 4y = -6$ and $2x - 4y = 18$, using elimination.

a. $(-3, 3)$

b. $(5, -3)$

c. $(3, -5)$

d. $(3, -3)$

31 [Intermediate] • Target: A well-designed word problem: ticket sales at a community theater

Solve the system: $2x + 4y = 40$ and $2x - 4y = -24$, using elimination.

a. $(8, 4)$

b. $(4, 6)$

c. $(4, 8)$

d. $(6, 8)$

32 [Intermediate] • Target: Multi-Step Linear Equations & Identity Conditions

Solve the system: $3x + y = -17$ and $3x - y = -37$, using elimination.

a. $(10, -9)$

b. $(-7, 10)$

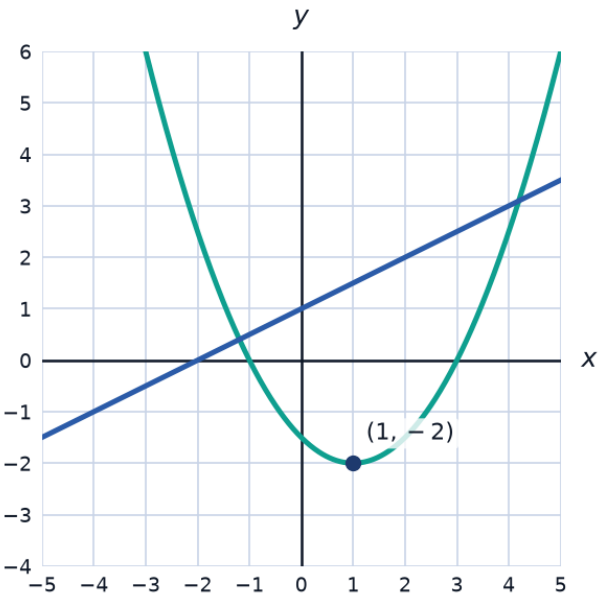
c. $(-9, 10)$

d. $(-9, 8)$

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33 [Intermediate] • Target: Linear Inequalities & Feasible Regions

Solve the system: $x - 5y = -48$ and $x + 5y = 52$, using elimination.



a. $(2, 8)$

b. $(4, 10)$

c. $(2, 10)$

d. $(10, 2)$

34 [Intermediate] • Target: Systems of Linear Equations

Solve the system: $5x - y = -24$ and $5x + y = -6$, using elimination.

a. $(-1, 9)$

b. $(-3, 7)$

c. $(-3, 9)$

d. $(9, -3)$

35 [Intermediate] • Target: Rate, Unit Conversions & Real-World Modeling

Solve $|2x - 12| = 26$.

a. 19

b. $-7, 19$

c. $7, -19$

d. $-7, 20$

36 [Intermediate] • Target: Solving linear equations

Solve $|2x - 7| = 21$.

a. 14

b. $7, -14$

c. $-7, 14$

d. $-7, 15$

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37 [Intermediate] • Target: Linear inequalities

Solve $-4x - 2 \geq -42$.

a. $x \geq 10$

b. $x \leq 10$

c. $x \leq 12$

d. $x \leq 8$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Slope and equations of lines • Cross-Lesson: 5.01 Statistics & Probability

Solve $-7x - 3 \geq 53$.

a. $x \leq -8$

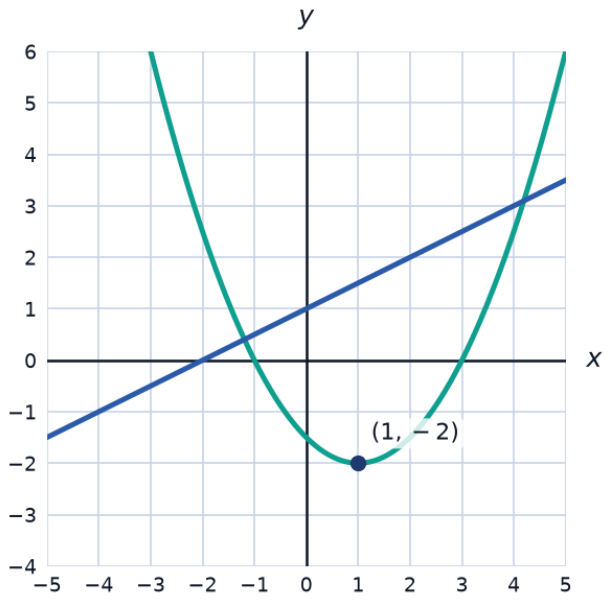
b. $x \leq -10$

c. $x \leq -6$

d. $x \geq -8$

39 [Challenging] • Target: Systems of equations by substitution • Cross-Lesson: 6.01 Geometry

Two numbers have a sum of 114 and a difference of 16. Find the larger number.



a. 63

b. 49

c. 67

d. 65

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40 [Challenging] • Target: Systems of equations by elimination • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Two numbers have a sum of 147 and a difference of 15. Find the larger number.

a. 81

b. 79

c. 66

d. 83

41 [Challenging] • Target: Multi-Step Linear Equations & Identity Conditions • Cross-Lesson: 8.01 Exponentials & Logarithms

If $5(2x - 1) - 3(x + 4) = 4(x - 3) + 7$, what is the value of x?

a. 4

b. 6

c. 8

d. 12

42 [Challenging] • Target: Multi-Step Linear Equations & Identity Conditions • Cross-Lesson: 2.01 Quadratics & Polynomials

If $\frac{2x - 5}{3} = \frac{x + 4}{2}$, what is the value of $3x + 1$?

a. 22

b. 64

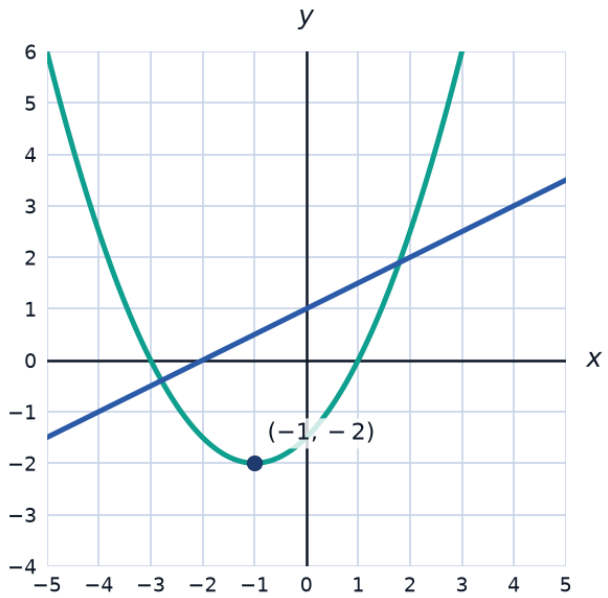
c. 67

d. 70

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43 [Challenging] • Target: Linear Inequalities & Feasible Regions • Cross-Lesson: 3.01 Functions & Graphs

If $-3x + 7 \geq 22$, what is the greatest possible value of x?



a. -5

b. -3

c. 5

d. 15

44 [Challenging] • Target: Systems of Linear Equations • Cross-Lesson: 4.01 Ratios, Rates & Data

For what value of k does the system $kx + 4y = 8$ and $3x + 2y = 5$ have no solution?

a. 3

b. 6

c. -6

d. 12

45 [Challenging] • Target: Rate, Unit Conversions & Real-World Modeling • Cross-Lesson: 5.01 Statistics & Probability

A car uses 8 liters of fuel per 100 kilometers. How many liters of fuel are required for a trip of 350 kilometers?

a. 24

b. 26

c. 28

d. 32

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